

INTRODUCTION

Face recognition technology has become an important biometric authentication method due to its convenience, speed, and ease of use. It is widely adopted in smartphones, banking services, attendance systems, e-commerce applications, surveillance, and secure access control. Although face recognition systems can accurately identify individuals, they are vulnerable to presentation attacks, where attackers attempt to deceive the system using printed photographs, replayed videos, mobile displays, or 3D masks.

Face anti-spoofing, also known as face liveness detection, is designed to prevent such attacks by determining whether the presented face belongs to a real person. Traditional approaches often rely on handcrafted features or additional hardware such as infrared or depth cameras. These methods increase system complexity and deployment cost, making them less suitable for practical real-world applications.

Recent advances in deep learning have significantly improved the performance of face anti-spoofing systems. Convolutional Neural Networks (CNNs) can automatically learn discriminative facial features that distinguish genuine faces from spoof attacks. Among these models, EfficientNet-B0 offers a good balance between accuracy and computational efficiency, making it suitable for real-time applications.

In this project, a real-time face anti-spoofing system is developed using a standard RGB webcam, MediaPipe or YOLO Face for face detection, EfficientNet-B0 for feature extraction and classification, and a React-Flask web application for user interaction. The current implementation focuses on detecting whether a face is Live or Spoof. Future enhancements will extend the system to identify specific spoof attack types such as printed photo attacks, replay video attacks, mobile screen replay attacks, and 3D mask attacks.

The proposed system aims to improve biometric authentication security by providing an accurate, lightweight, and practical face anti-spoofing solution suitable for real-world deployment.